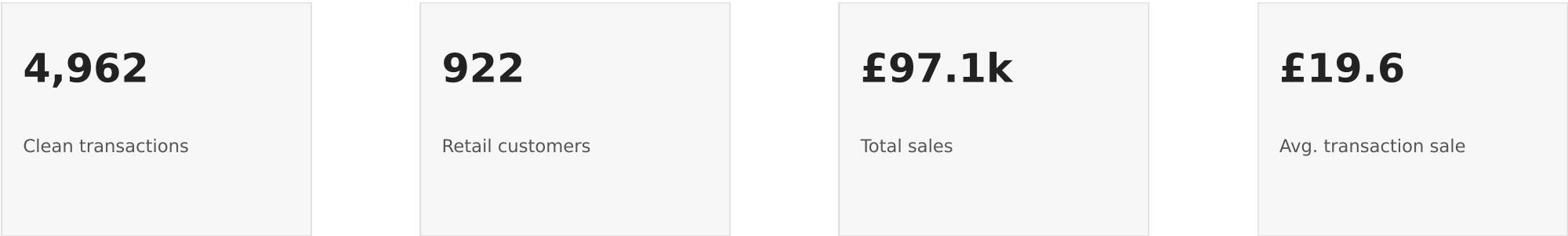
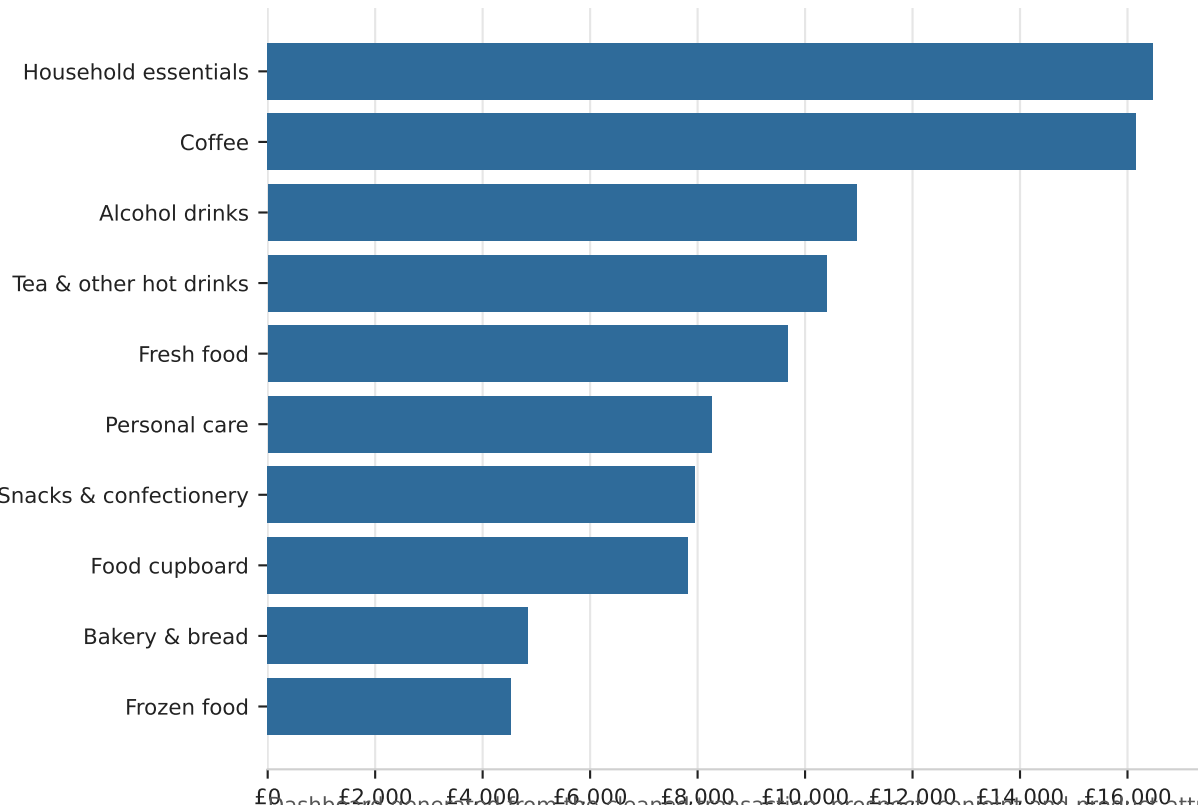


Executive launch recommendation

A data-led route from retail transactions to product markup, target segment and private-label coffee concept.



Revenue by category: coffee is a core commercial category



Dashboard generated from the cleaned transaction, prospect, conjoint and product attribute data. PNA/non-product rows are removed and missing product category values are treated as Unknown before filtering.

What the dashboard concludes

Target

Primary: Cluster 3 / High-value loyal customers. Secondary: Cluster 2 / mainstream active shoppers.

Product

Capsule coffee, dark roast, 100% Arabica blend, credible sustainability claim.

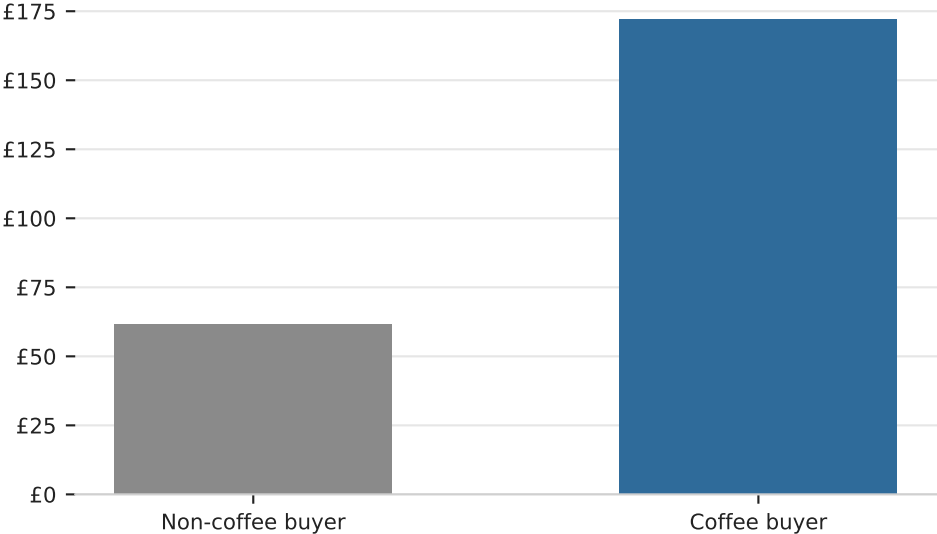
Price

£5 launch price: preserves value perception while avoiding the utility drop at £6.

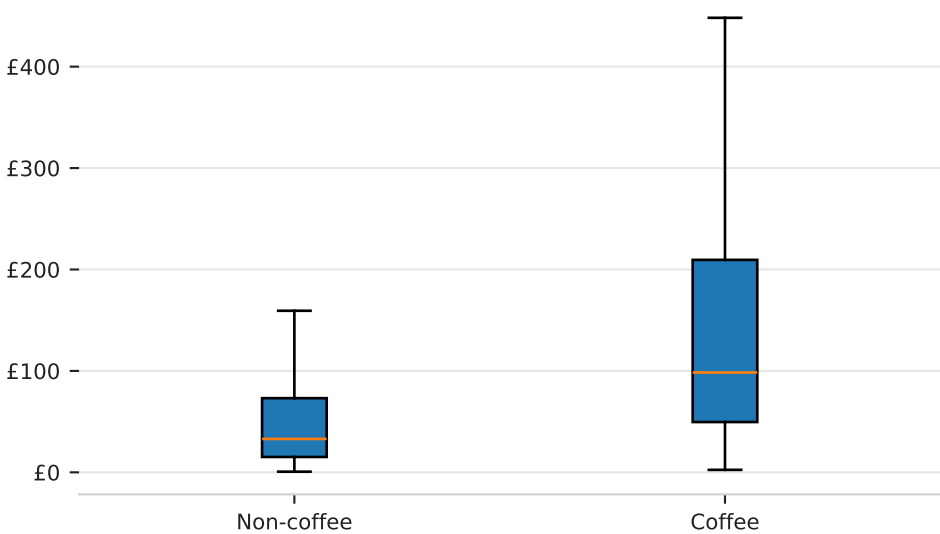
Expected result

Predicted share in the tested choice set: 31.8%.

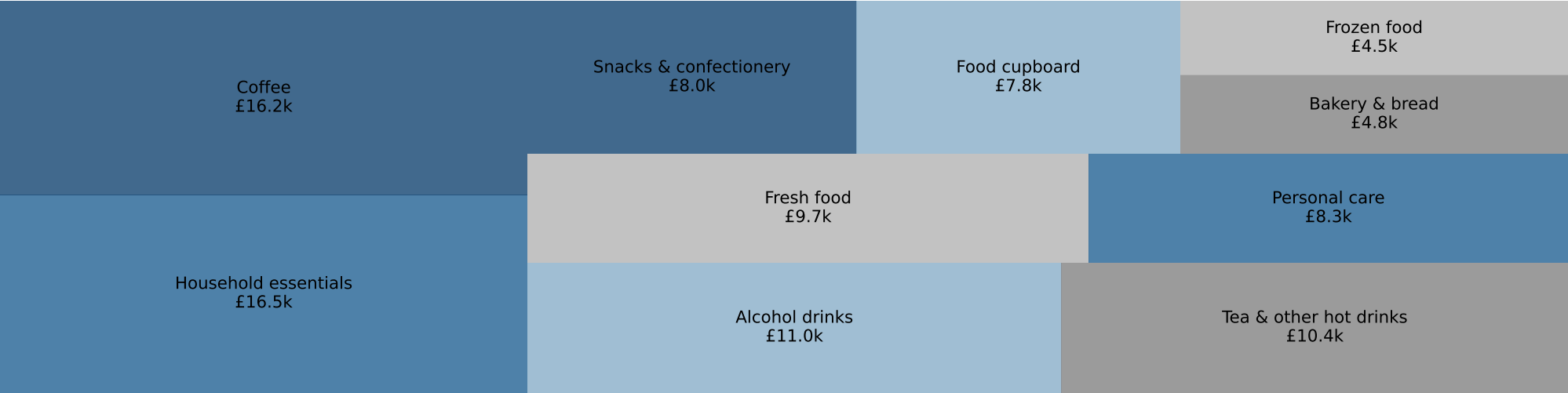
Average customer spend



Spend distribution, outliers hidden for readability



Category sales treemap

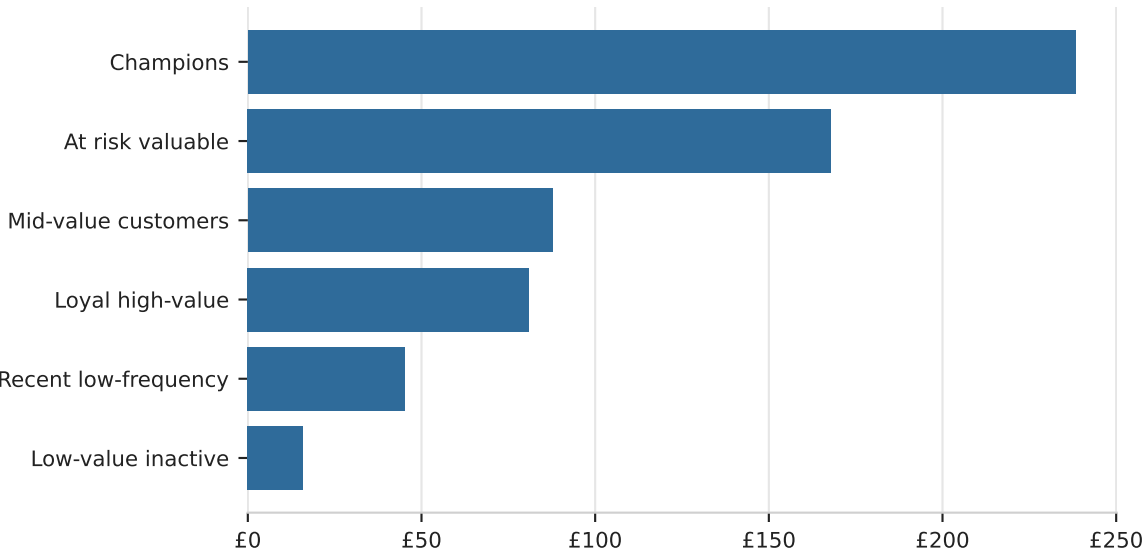


Use coffee-buyer status as an early behavioural signal: the category already captures customers with higher total value.

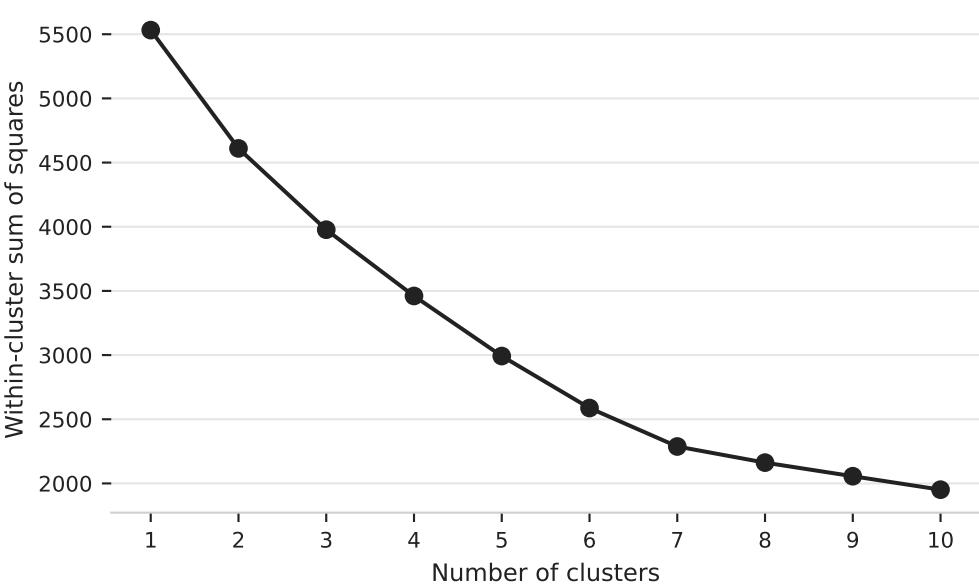
Segmentation: who should the launch target?

RFM identifies value tiers; K-means adds behavioural and demographic depth.

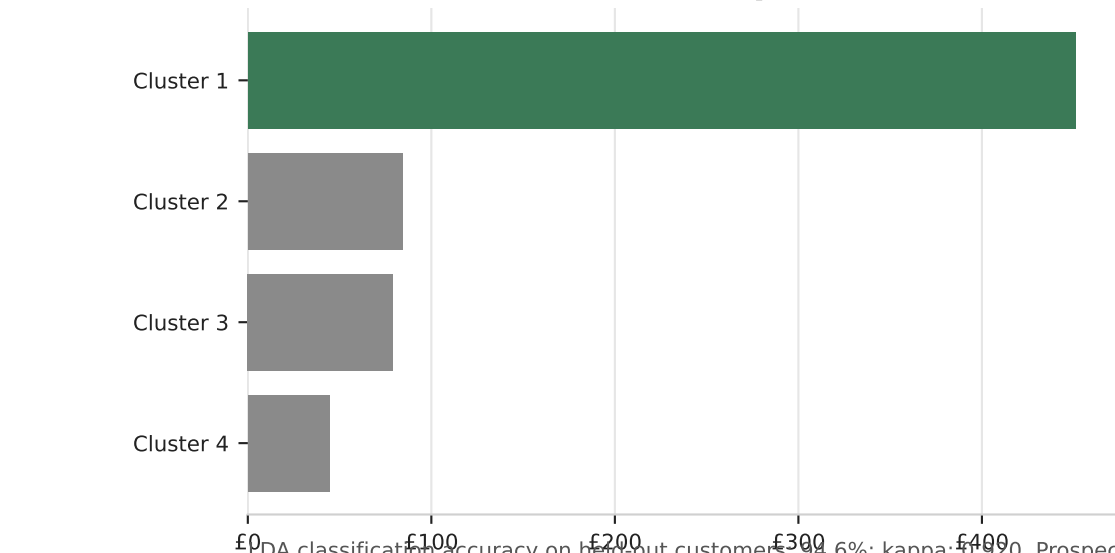
Average monetary value by RFM segment



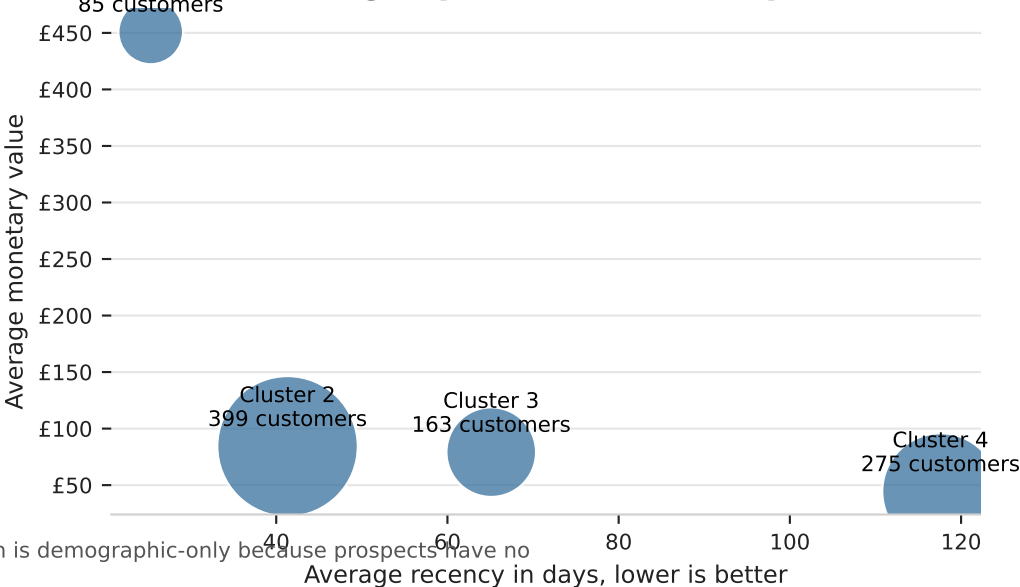
Elbow method supports a compact 4-cluster solution



Cluster value profile



Target prioritisation map

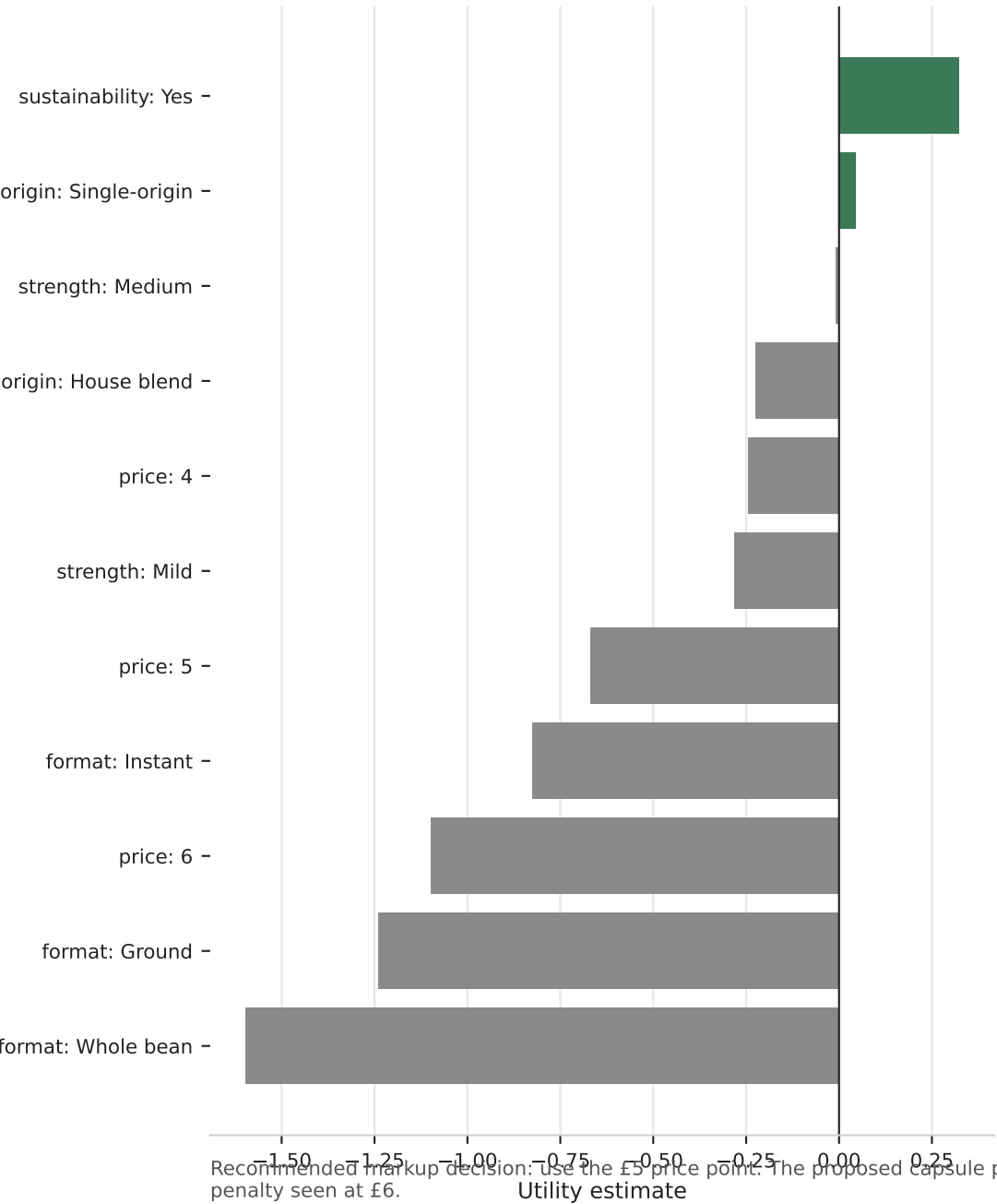


LDA classification accuracy on held-out customers: 94.6%; kappa: 0.920. Prospect classification is demographic-only because prospects have no transaction history.

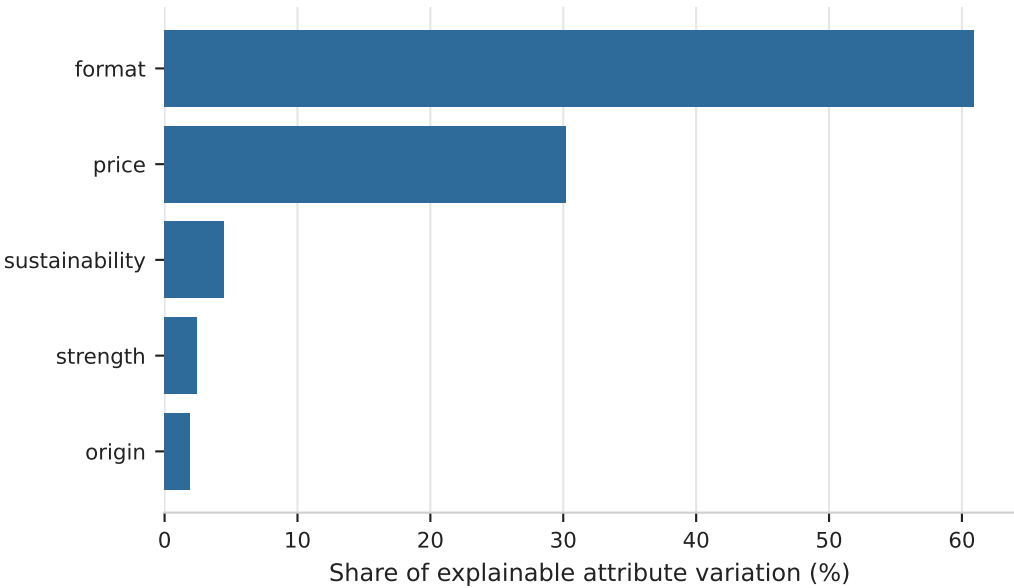
Conjoint: what product should be launched?

Preference is driven primarily by format and price, with sustainability adding measurable value.

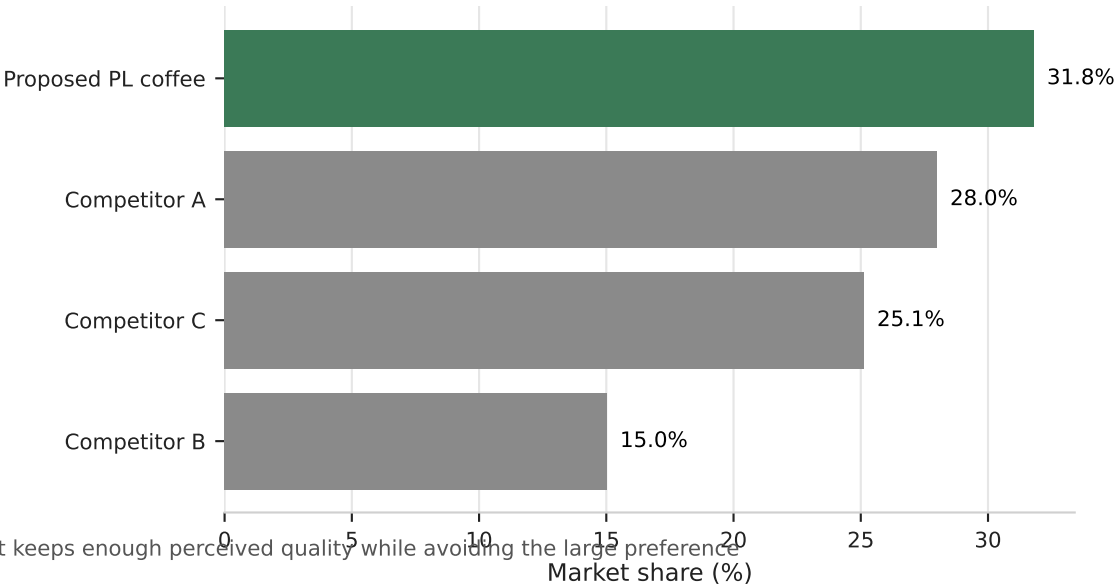
Part-worth utilities versus reference levels



Attribute importance



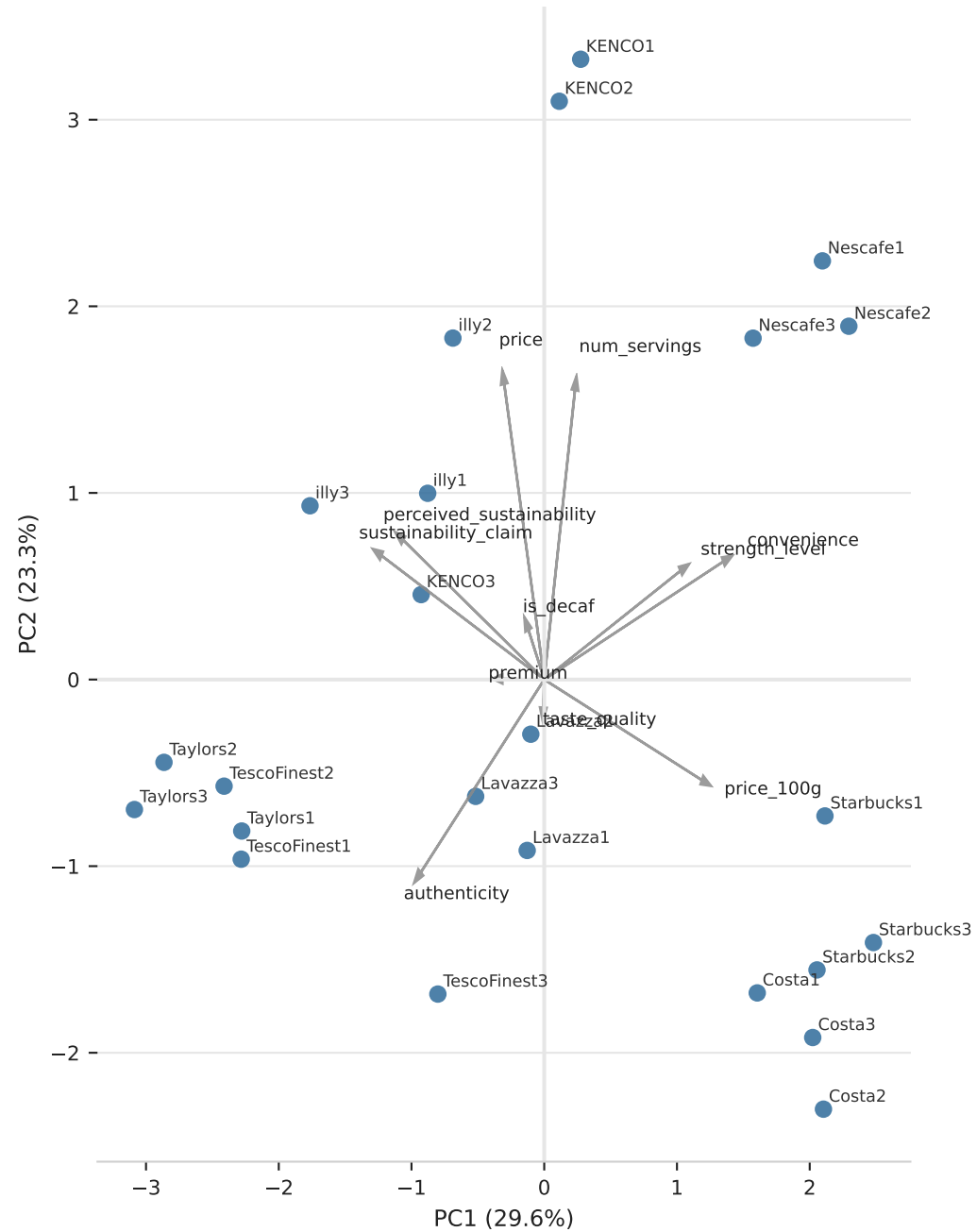
Simulated market share in test choice set



Positioning and 4P launch plan

PCA explains 53.0% of product-attribute variation across PC1 and PC2.

PCA perceptual map



4P strategy

Product Dark roast 100% Arabica capsule coffee with a credible sustainability claim.

Price Launch at £5: a mid-market value point with less utility loss than £6.

Place Coffee aisle, premium own-label bay, and online grocery search/category placements.

Promotion Loyalty-card trial for high-value loyal customers; digital expansion to mainstream active shoppers.

Final product markup

Private-label capsule coffee | £5 | dark roast | 100% Arabica | sustainability claim

Cluster profile

cluster	customer_size	avg_recalls	avg_freq	avg_mor	avg_incc	avg_age	avg_household_size
1	85	25.34	10.39	450.84	60.35	58.15	2.39
2	399	41.32	3.17	84.31	48.07	50.59	3.1
3	163	65.12	3.05	79.19	150.8	52.58	2.89
4	275	117.6	1.64	44.35	50.64	61.94	2.15

Predicted prospect membership

cluster	prospects
2	3047
3	1016
4	937

Conjoint importance

attribute	importance
format	60.93
price	30.23
sustainability	4.46
strength	2.47
origin	1.91

Choice-set simulation

product	predicted_rating	market_share
Proposed PL coffee	5.57	31.82
Competitor A	5.44	28.01
Competitor B	4.82	15.04
Competitor C	5.33	25.14